

Lumen — Portfolio Documentation

AI-Based Web Accessibility Checker

Repository: github.com/balisikh/ai-based-web-accessibility-checker · Local: <http://localhost:4376>

Assistive findings only — not a legal accessibility certificate or formal conformance audit.

Overview

Lumen scans a public URL with Playwright + axe-core, returns a score, severities, rule help, optional AI tips, and JSON export.

1. Paste a public website URL
2. Headless Chrome renders the page
3. axe-core checks WCAG A/AA-oriented rules
4. Score, severities, and issue details
5. Optional AI tips; JSON export

UI screenshots

L Lumen

CheckerBatch resultsLightDarkSystem

Lumen

WCAG-ORIENTED · LIVE PLAYWRIGHT + AXE

Accessibility checker

Paste a public URL. Get automated accessibility findings, a clear score, and links to rule help.

Website URL

https://example.com

Check accessibility

Try

example.com

W3C bad demo

Public http or https URLs only — private and local addresses are blocked.

PORTFOLIO SNAPSHOT

28 sites tested

8 Pass · 20 Fail

366 issues · resync 2026-07-28

Open batch dashboard →

Pass vs Fail guidance

See severity totals and recommended actions for each public URL in the portfolio test set.

Pass rule (websites)

Score ≥ 85 with zero critical issues. Sites can still have moderate or minor findings — triage those like Fail follow-ups.

Assistive only

Automated axe findings help you improve pages; they are not a legal WCAG certificate or full manual audit.

How to use Lumen

Step-by-step from URL to report.

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Step-by-step from URL to report.

1

Enter a public URL
Paste an http or https link in the field above (for example https://example.com), or pick a Try example.

2

Check accessibility
Click Check accessibility to scan that single page. Localhost and private network addresses are blocked.

3

Wait for progress
Follow the status steps while Lumen fetches the page, runs rules, optional AI tips (when enabled), and calculates a score. Most public pages finish within about a minute.

4

Review findings
Check the score and severity counts. Select an issue to see WCAG mapping, HTML snippet, rule help, and AI guidance when enabled on this server. Filter by severity if the list is long.

5

Export JSON (optional)
On the results screen, use Export JSON to download the report for tickets, docs, or automation.

How Lumen checks a page

AI tips off

1

Public websites
Use http or https links that anyone can open on the internet.
✓ https://example.com ✗ localhost or 127.0.0.1 ✗ 192.168.x.x

2

Real browser scan
Automated Chrome loads the page in the background, like a real visitor.

3

Accessibility rules
Industry-standard checks (axe-core) for common WCAG Level A and AA requirements.

4

Optional AI tips
Not enabled on this Lumen server — you still get rule findings, WCAG mapping, and a score.

Website batch results

+ What URLs can I use?

+ What are the limits of a scan?

+ How is the score calculated? What does Pass / Fail mean?

+ What are AI tips? Why are they off?

+ Why did my scan fail or say rate limited?

+ What happens to my URL and scan data?

Batch results — 28 sites (overview)

L Lumen

CheckerBatch resultsLightDarkSystem

PORTFOLIO TEST SET

Run a live scan

Website batch results

Static snapshot (last full live resync 2026-07-28) — loads instantly and does not run scans. Live checks use the accessibility checker.

TESTED28

PASSED84 clean - 4 with issues

FAILED20

TOTAL ISSUES366

CRITICAL126

SERIOUS30

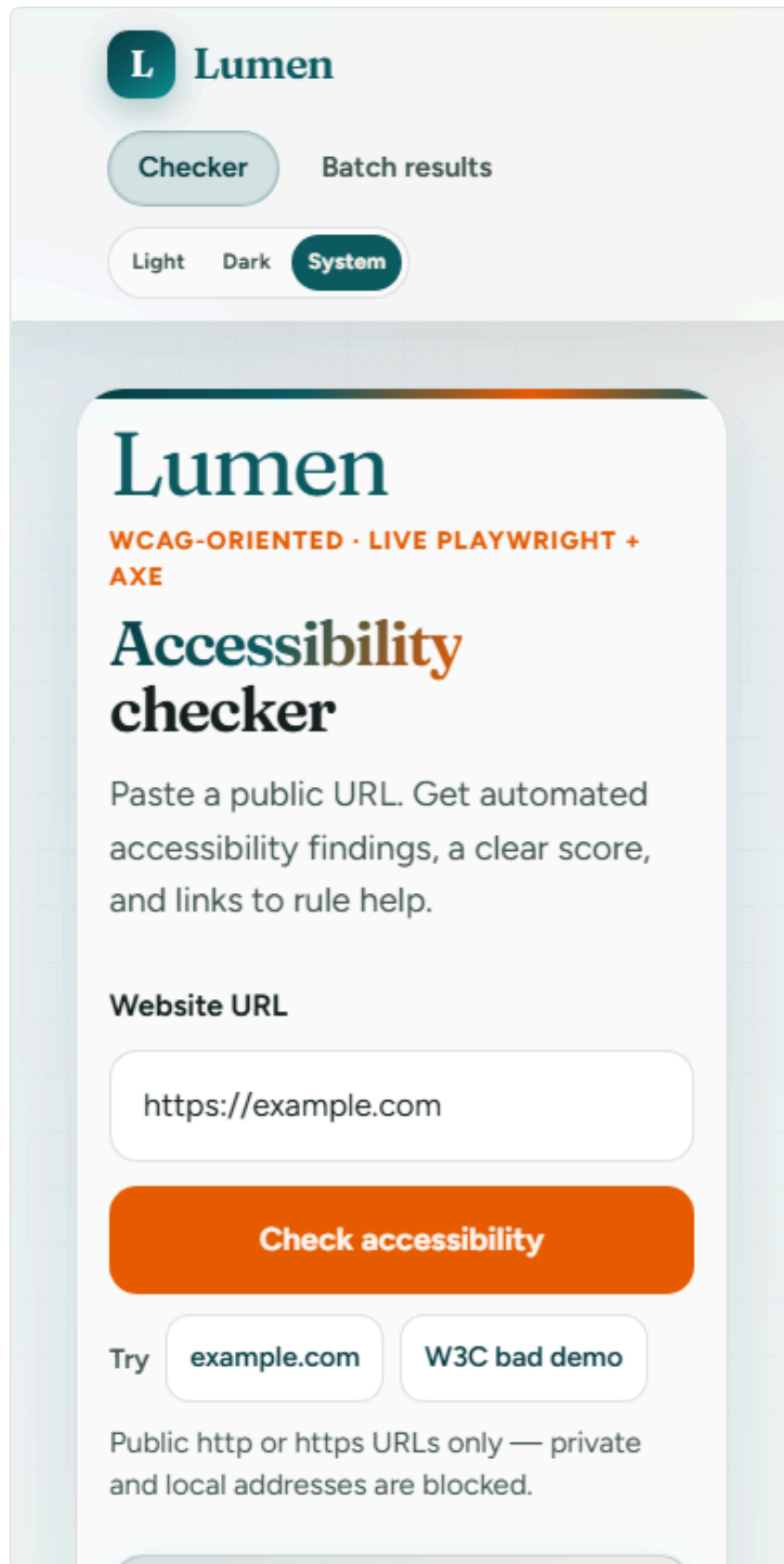
MODERATE189

MINOR21

Pass: score ≥ 85 and critical = 0. Fail: otherwise. Pass and Fail rows both use Notes & recommendations when there are issues to triage. Source: TEST_RESULTS.md in the repo.

#	WEBSITE	SCORE	CRITICAL	SERIOUS	MODERATE	MINOR	TOTAL	RESULT	NOTES & RECOMMENDATIONS
1	Google UK BENCHMARK PASSCLEAN SCAN	100	0	0	0	0	0	PASS	MAINTAIN PASS NOTE Perfect automated score (0 issues). Use as a clean baseline when comparing other scans. Re-scan after major Google UI changes — automated results can shift. Keep manual spot-checks; Pass is not a formal WCAG certificate.
2	YouTube CRITICAL BLOCKER	0	4	0	0	0	4	FAIL	4 critical issues 4 critical issues; score 0 (need ≥ 85) 4 issues — work through recommendations below RECOMMENDED ACTIONS Fix 'aria-allowed-attr' on main nav links (remove or correct invalid ARIA). Re-scan in Lumen and use Rule help on each critical issue. Target zero critical issues, then re-check score ≥ 85. Re-scan in Lumen for live rule detail.

3	BBC weather Southall	54	1	0	3	0	4	FAIL	1 critical issue 1 critical issue; score 54 (need ≥ 85) 4 issues — work through recommendations below	RECOMMENDED ACTIONS Fix critical 'aria-required-attr' first. Resolve 'aria-hidden-focus' and 'color-contrast' (serious) on weather UI. Re-scan after BBC deploys changes (page content changes often). Re-scan in Lumen for live rule detail.
4	BBC iPlayer	86	0	0	2	0	2	PASS	2 issues — batch Pass; triage below (keep critical at 0)	RECOMMENDED ACTIONS NOTE Pass with 2 moderate issues (score 86). Open each moderate issue in Lumen Rule help and fix when convenient. Re-scan to confirm critical stays at 0 and score stays ≥ 85. Prioritize any new critical findings before release. Still a batch Pass — re-scan in Lumen after fixes so critical stays 0
5	BBC News	93	0	0	1	0	1	PASS	1 issue — batch Pass; triage below (keep critical at 0)	RECOMMENDED ACTIONS NOTE Pass with 1 moderate issue (score 93). Triage the moderate finding; low effort may lift score further. Re-scan after BBC template or component updates. Export JSON if tracking regressions in a ticket. Still a batch Pass — re-scan in Lumen after fixes so critical stays 0
6	Disney+ UK	100	0	0	0	0	0	PASS	CLEAN SCAN	MAINTAIN PASS NOTE Clean scan — 100 score, zero issues. Re-scan periodically; streaming sites update often Use as a Pass reference alongside Google UK / Maps. Still validate key flows manually (keyboard, captions etc.).
7	GitHub balisikh	100	0	0	0	0	0	PASS	CLEAN SCAN	MAINTAIN PASS NOTE Profile page Re-scan if GitHub changes profile markup or ARIA patterns. Scan other URLs separately in testing org/repo pages. Maintain zero critical on re-scan to keep batch Pass.
8	ChatGPT	50	1	1	1	1	4	FAIL	1 critical issue 1 critical issue; score 50 (need ≥ 85) 4 issues — work through recommendations below	RECOMMENDED ACTIONS Correct unsupported ARIA ('aria-allowed-attr', critical). Improve text contrast ('color-contrast', serious). Re-scan; export JSON to track before/after. Re-scan in Lumen for live rule detail.
9	Spotify Web Player	0	82	0	2	16	100	FAIL	82 critical issues 82 critical issues; score 0 (need ≥ 85) 100 issues — work through recommendations below	RECOMMENDED ACTIONS NOTE Re-scan: app shell / login — confirm in UI Confirm URL and cookies (open.spotify.com vs marketing landing). Triage 82 critical findings — fix invalid ARIA/roles first. Re-scan after stable page load; compare to prior Pass (100/0). Re-scan in Lumen for live rule



Layout (light mode)

- **Header:** Lumen logo, Checker / Batch results, Light / Dark / System
- **Background:** Soft grey gradient + grid pattern
- **Card:** White panel with teal gradient strip on top
- **Home CTA:** Orange Check accessibility button

- **Batch CTA:** Teal Run a live scan button

Batch snapshot (2026-07-30)

Metric	Value
Websites tested	28
Passed	8 (4 clean · 4 with issues)
Failed	20
Total issues	374

Pass rule: Score ≥ 85 and Critical = 0.

Passed sites: Google UK, BBC iPlayer, BBC News, Disney+ UK, GitHub (balisikh), Google Maps, Wikipedia, example.com

MVP features

Area	Status
Scan UI, Playwright + axe	Done
Rate limiting, persistence, JSON export	Done
Optional AI tips, anonymous use	Done
PDF export / accounts / crawl	Not yet

Stack

- Next.js (React) + TypeScript
- Playwright (Chrome) + axe-core
- Postgres or PGlite
- Optional OpenAI-compatible AI tips

Local development note — plain-text UI incident (Aug 2026)

During local testing the app briefly showed **unstyled plain text** (e.g. nav labels run together). The design was not removed — **CSS failed to load**.

Symptom: White page, black text; HTML OK but stylesheet returned **500** from a stale/corrupt .next build on port 4376.

Cause: On low-memory Windows, Turbopack could crash compiling `globals.css`, leaving a broken build that still served HTML.

Fix (commit 541ec96):

- `dev / build` default to **webpack**
- `web/scripts/fix-styles.ps1` — stop server, delete `.next`, rebuild, restart
- Troubleshooting in `web/README.md` — confirm CSS status **200** in DevTools → Network

Resolution: After clean rebuild, UI matched screenshots again (header, gradient, card, batch table). Images in this PDF show the intended layout.

Generated 2026-08-04 from docs/screenshots and project README.